

AA-K1 User Manual

Version 1.2

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Manufactured & Written
By

Libertron Co., Ltd

Republic of Korea

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➤ Revision History

Version	Date	Description
1.0	2026-02-24	1 st Initial Version.
1.1	2026-03-26	Revises the description about CAN Interface ID, CAN Message ID
1.2	2026-04-15	Troubleshooting 예 9.1 Actuator Not Detected(Connection Issue) 추가

1. Introduction

1.1. Purpose of This Manual

This manual provides instructions for installation, configuration, and communication testing, and debugging of the AA-K1 robotic system.

1.2. Intended Users

This manual is intended for researchers, professors, and students with fundamental knowledge of robotics systems and Linux-based environments.

2. Safety Information

⚠ General Safety Notice

- Install the main unit on a stable surface before operation.
- Always secure the main unit properly before motion testing.
- Keep hands and objects away from moving joints.
- Disconnect power before modifying hardware connections.
- At least two people are required when lifting the main unit out of the box.
- Ensure the Emergency Stop button is always accessible.
- When installing main unit, securely fasten it with screws or clamps to a flat and stable surface.
- Failure to follow these instructions may result in equipment damage or personal injury.



3. Getting Started

3.1. Unboxing & Handling

The box contains Main Unit of dual robot arm and Accessory Box.

1) Foam Removal

Remove all foam materials until the entire AA-K1 Main Unit is fully visible.

Carefully inspect the robot to ensure no packaging materials remain attached to the joints or gripper.

2) Lifting the Main Unit

- The main unit must be lifted by at least two people.
- Before lifting, you should read and refer to "4.1 Placement and Securing of the Main Unit"
- Lift main unit by holding the main frame and arms simultaneously.
- Do not lift the main unit using only the arm links, wrist, or gripper.
- Make sure all movable parts are clear and unobstructed before raising the unit.
- Failure to follow these instructions may result in equipment damage.

3) Accessory Box Handling

Lift out the Accessory Box located beneath the main unit. The Accessory Box contains:

- Power Unit
- Communication (USB 2.0/USB-B) cable
- Power cable
- Optional Camera components (if purchased)

3.2. Package Contents

After unpacking, verify that all components listed below are included.

If any items are missing or damaged, contact Libertron Co., Ltd. before proceeding with installation.

Item	Qty	Description
AA-K1 Main Unit	1	Robotic manipulator
Accessory Box	1	Contains Power Unit, cables, and optional components
Power Unit	1	Integrated power supply and USB-to-CAN FD converter
Power Cable	1	AC power cable
USB 2.0 Cable (A to B)	1	PC communication cable
Chest Camera*	1	Factory-installed (if selected)
Arm Cameras*	2	Supplied separately for user installation (if selected)
Arm Camera Mounting Kit*	2	Includes mounting brackets and screws (if selected)

* Optional items. Included only if selected at the time of purchase.



⚠ Inspection Notice

- Inspect all items for visible damage before installation.
- Do not apply power to the system if any components appear damaged.
- Cameras and Mounting Kit are optional items.

4. Hardware Installation

4.1. Placement and Securing of the Main Unit

Place the main unit on a flat and stable surface with sufficient clearance for its full range of motion.

Ensure that:

- The installation surface can support the weight of the main unit.
- The surface is level and free from vibration.
- There are no obstacles within the operating area of the arm.
- The main unit is positioned securely before operation.

During operation, dynamic motion may generate **vibration or shifting forces**.

To prevent the robot from **tipping forward** or unintended movement, securely fasten the main unit to the installation surface using **clamps, bolts, or similar fixtures**.

Do not rely solely on the weight of the main unit for stability.

Failure to properly secure the unit may result in equipment damage or personal injury.

4.2. Camera Options

Two optional camera configurations are available:

- Chest Camera
- Arm Cameras (Left and Right)

1) Chest Camera

The Chest Camera is factory-installed.

No user installation is required.

2) Arm Cameras

Arm Cameras are supplied separately inside the Accessory Box.

If included, the user must install them using the provided mounting brackets and screws.

After installation, ensure that:

- The cameras are firmly secure.

- Cables are properly routed.
- No cable interferes with joint motion.
- Improper installation may restrict movement or cause damage during operation.

4.3. Power Unit Installation and PC Connection

1) Power Unit Overview

The Power Unit integrates:

- Internal power supply
- USB-to-CAN FD converter
- Top-mounted Emergency Stop (E-Stop) button

cf. Pressing the E-Stop immediately cuts power to the robot.

2) AC Power Connection

Connect the power cable to the Power Unit and plug it into a suitable AC outlet.

Ensure the power source is stable before proceeding.

3) Robot Power Connection

The power cable for the robot is **pre-attached to the main unit**.

Connect this pre-installed cable from the main unit to the output port of the Power Unit.

Ensure the connector is fully inserted and secured before powering on the system.

4) PC Connection (USB-B connection)

Connect USB-B port of the Power Unit to the PC via the USB 2.0 cable.

The integrated USB-to-CAN FD converter enables communication between the main unit and the computer.

Use a PC running Ubuntu 22.04 or 24.04 for compatibility.

⚠ Safety Reminder

Do not power on the system until all connections are properly secured.

5. Software Installation

5.1. System Requirements

Before installing the software, ensure that your system meets the following requirements:

- Ubuntu 22.04 or Ubuntu 24.04
- Linux system with SocketCAN support
- CMake 3.22+
- C++17 compiler
- Internet connection for package installation

The USB-to-CAN FD converter is integrated into the Power Unit and CAN FD will be recognized as a network interface in Linux.

5.2. Installing OpenArm Software Packages

```
</> Bash
sudo apt install -y software-properties-common
sudo add-apt-repository -y ppa:openarm/main
sudo apt update
sudo apt install -y \
    can-utils \
    iproute2 \
    libopenarm-can-dev \
    openarm-can-utils
```

5.3. Verifying Installation

```
</> Bash
which candump
which cansend
```

If the installation was successful, the system should return valid paths (e.g., /usr/bin/candump or /usr/bin/cansend).

If no path is returned, reinstall the **can-utils** package.

5.4. Download the CAN Library

```
</> Bash
wget
https://github.com/enactic/openarm_can/releases/download/1.2.8/openarm-
can-1.2.8.tar.gz

tar xf openarm-can-1.2.8.tar.gz
cd openarm-can-1.2.8
```

⚠ Note.

Version **1.2.8** is the current stable release as of this writing. It is recommended to check the for the **latest version** and update the version strings in the commands above accordingly.

6. CAN Interface Configuration

6.1. Identifying the CAN Interface

Run the following command to check the CAN Interfaces

```
</> Bash
```

```
ip link show
```

Ensure that two CAN interfaces (e.g., can0 and can1) are displayed in the output.

If ip link show does not display the expected CAN Interfaces, please refer to "9.1 CAN Interface Not Detected".

⚠ Important Notice

The CAN Interface configuration is **not persistent**.

This configuration must be repeated each time the PC is restarted, or the USB connection is re-established.

6.2. Configuring the CAN FD Interface

Required settings:

- **Nominal bitrate: 1 Mbps**
- **Data bitrate: 5 Mbps**
- **CAN FD mode enabled**

CAN FD can be configured using the following two methods:

1) Recommendation

The OpenArm CAN library provides a simplified setup command. Configure each identified Interface (e.g., can0 and can1)

Example for can0 and can1:

```
</> Bash
```

```
openarm-can-configure-socketcan can0 -fd -b 1000000 -d 5000000
```

```
openarm-can-configure-socketcan can1 -fd -b 1000000 -d 5000000
```

2) Manual Configuration

If you prefer manual setup, use the following commands for each Interface.

Example for can0:

```
</> Bash
```

```
sudo ip link set can0 down
```

```
sudo ip link set can0 type can bitrate 1000000 dbitrate 5000000 fd on
```

```
sudo ip link set can0 up
```

⚠ Note on Interface Identification

The number of identified Interfaces depends on the number of **USB-B connections** established (as described in Section 4.3):

Single Connection: 2 interfaces (e.g., can0, can1) will be displayed.

Dual Connection: 4 interfaces (e.g., can0 to can3) will be displayed.

Cf. For example, a single Follower Arm Set requires one USB-B connection via a USB 2.0 cable, while using both the Leader and Follower Arm Sets requires two USB-B connections via two USB 2.0 cables.

7. PC and Main Unit Communication Test

7.1. Default CAN Interface ID Assignment Policy

1) Basic Rules for Assignment at the PC Level

Although both devices are physically connected at the same time, the system (PC) assigns CAN Interface IDs according to a predefined rule:

- The **Right Arm** is always assigned the **lower CAN Interface ID**.
- The **Left Arm** is assigned to the **higher CAN Interface ID**.

Ex) Follower-Right-Arm: can0, Follower-Left-Arm: can1, or Follower-Right-Arm: can1, Follower-Left-Arm: can2. ...

When identifying devices and interfaces on the PC side, the lower CAN Interface ID must be recognized as the Right Arm, and the higher CAN Interface ID as the Left Arm.

2) Use case of Two Sets (Leader and Follower Configuration)

When connecting two sets of Main Unit to the PC via USB 2.0 cables, the CAN Interface and connection is established sequentially like below.

- The set connected **first** is assigned the **lower CAN Interface ID** range.
- The set connected **second** is assigned the **higher CAN Interface ID** range.

Ex) Follower-Right-Arm: can0, Follower-Left-Arm: can1, Leader-Right-Arm: can2, Leader-Left-Arm: can3 or Leader-Right-Arm: can0, Leader-Left-Arm: can1, Follower-Right-Arm: can2, Follower-Left-Arm: can3. ...

The allocation of the lower CAN Interface ID range to either the Leader or the Follower is entirely at the user's discretion.

The system is designed to operate normally regardless of which role (Leader or Follower) receives the lower CAN Interface IDs.

7.2. Default CAN Message ID Configuration for each Joint

Joint	Name	Sender ID	Receiver ID
1	Shoulder Pan	0x01	0x11
2	Shoulder Lift	0x02	0x12
3	Arm Roll	0x03	0x13
4	Elbow Pitch	0x04	0x14
5	Wrist Roll	0x05	0x15
6	Wrist Pitch	0x06	0x16
7	Wrist Yaw	0x07	0x17
8	Gripper	0x08	0x18

The Sender ID is used for device-to-master communication, while the Receiver ID is used for master-to-device communication.

- Sender ID: used when Device(Joint) to Master(PC) messaging
- Receiver ID: used when Master(PC) to Device(Joint) messaging

⚠ Note.

The **CAN Message ID assignments for each Joint(Actuator)** above are **pre-configured at the factory**.

Users are not required to configure these settings, and doing so is not recommended.

7.3. Monitoring CAN Traffic

To monitor traffic for a specific arm, use the assigned interface name (e.g., can0 or can1).

Example for Right Arm:

```
</> Bash  
candump can0
```

7.4. Sending a Test Command

The following examples use can0 as a reference. If your target arm is assigned to a different interface, replace can0 with the correct name (e.g., can1).

1) Enable Joint 1 actuator

```
</> Bash
```

```
cansend can0 001#FFFFFFFFFFFFFFFC
```

This command will turn **on** the LED on Joint 1 actuator.

2) Disable Joint 1 actuator

```
</> Bash
```

```
cansend can0 001#FFFFFFFFFFFFFFFD
```

This command will turn **off** the LED on Joint 1 actuator.

Safety Notice

Ensure the main unit is secured and the workspace is clear before sending commands.

Be prepared to press the Emergency Stop if unexpected motion occurs.

8. Zero-Setting Procedure

This procedure defines the steps to align the hardware and execute the command required to record the arm's **Zero Point**. The system will automatically move to calculate its position, concluding with a final **"Zero Set"** to save the reference coordinates.

8.1. Physical Alignment (Pre-Setting)

Before initiating the software command, manually set the device to its reference posture. This alignment provides the starting point for the automated sequence.

- **Arm Position:** All joints of the arm must be fully extended and straightened so that the arm is in a completely linear state.
- **Gripper Position:** Ensure the gripper is in a fully closed position.

cf. Precise alignment is not required.

8.2. CAN Interface ID (canport) Assignment Check

Before performing the Zero Setting, ensure that the CAN interface ID (canport) is correctly assigned to each arm in accordance with Section 7.1.

8.3. Zero Calibration and Zero Position Recording

Execute the zero-setting command. The device will follow an automated sequence to determine its zero point.

Example for Right Arm:

```
</> Bash
openarm-can-zero-position-calibration --canport can0 --arm_side right_arm
```

Cf. If your target arm is assigned to a different interface, replace can0 with the correct name (e.g., can1).

Example for Left Arm:

```
</> Bash
openarm-can-zero-position-calibration --canport can1 --arm_side left_arm
```

Cf. If your target arm is assigned to a different interface, replace can1 with the

correct name (e.g., can2).

- **Auto-Positioning:** Upon execution, the arm moves autonomously to detect mechanical limits and calculate orientation.
- **Zero Set & Recording:** As the last step, the system calls the "Zero Set" function and stores the coordinates as the Zero Point.
- The Zero Point value remains stored regardless of power on/off.

⚠ Safety Notice

Safety First! The robot will move automatically when calibration starts. Wear safety gear, clear the workspace, and be ready for the emergency stop. Do not disconnect power until the process is complete.

9. Troubleshooting

9.1. Actuator Not Detected (Connection Issue)

If a specific actuator is not detected while others are functioning normally, the issue is most likely related to a connection problem rather than a hardware failure.

- **Check cable connection:** Ensure the connector between the actuator and controller is properly connected.
- **Reconnect cable:** Disconnect and reconnect the connector securely and verify proper alignment.

⚠ **Note.** In most cases, this is not caused by actuator failure.

9.2. CAN Interface Not Detected

If **ip link show** does not display the expected CAN Interfaces, perform the following hardware checks.

- **Check Power:** Ensure the Power Unit is turned on and properly powered.
- **Reconnect USB:** Disconnect and reconnect the USB-B cable to refresh the connection.
- **Try Different Ports:** Connect to another USB port on your PC to rule out port failure.
- **Verify USB Connection:** Run `lsusb` to confirm if the system recognizes the USB-to-CAN FD converter hardware.

9.3. No CAN Data in candump

If `candump` shows no output despite a successful connection, check the software configuration.

- **Reconfigure Interface:** The configuration is lost after a reboot or reconnection. Repeat the setup process in Section 6.2.
- **Verify Baudrate:** Ensure the Nominal (1 Mbps) and Data (5 Mbps) baudrates match the hardware presets.
- **Check Interface Status:** Run `ip link show can0` and ensure the state is "UP". If it is "DOWN", the interface must be activated.

9.4. Permission Denied

SocketCAN configuration requires administrative privileges.

- Always use the `sudo` prefix when executing `ip link` or configuration commands.

9.5. Command Not Found

If the system does not recognize commands like `candump` or `cansend`, the necessary utilities may not be installed.

- Install the standard CAN tools by running.

```
</> Bash
```

```
sudo apt install can-utils
```

Reserved Page